

Software Engineering

Assignment 05: UI Testing with XPath and CSS Selectors using Selenium

Total Marks: 40

Objectives:

- Learn and apply XPath and CSS Selectors.
- Use Selenium for UI testing automation.
- Practice locating UI elements through automation scripts.

Part A: Theoretical Understanding (10 Marks)

Answer the following questions **on paper** (handwritten), **scan** your pages and attach as a **PDF file** in your submission.

1. **What is Selenium and why is it used in automated testing?**
2. **Explain the difference between absolute XPath, relative XPath, and CSS selectors. Provide examples.**
3. **Compare XPath and CSS Selectors in terms of:**
 - Readability
 - Performance
 - Flexibility

Each answer should be 150–200 words, written neatly, and scanned clearly.

Part B: Practical Task – UI Testing Using Selenium (20 Marks)

Scenario:

You are testing a login and registration page for a sample web app.

Tasks:

1. Use or create a simple HTML file with:
 - Login form: username, password, login button
 - Registration form: name, email, password, confirm password, register button
2. Write **four Python scripts** using Selenium:
 - `test_login_xpath.py`
 - `test_login_css.py`
 - `test_register_xpath.py`
 - `test_register_css.py`
3. Use **absolute XPath, relative XPath, and CSS selectors** for element location.
4. Add **clear code comments** for each locator used.

Submission Must Include:

- All Python code files
- The HTML file (if created)
- `README.md` (how to run the scripts)

Attach Results:

- If scripts generate output → take screenshots of browser and console
- If the task involves form interaction without visible results → attach **step-by-step screenshots** of each script execution.

Part C: XPath & CSS Selector Tasks (10 Marks)

Write the correct XPath or CSS Selector (in a separate `.txt` or `.md` file) for:

1. Select an input field with id `username`.
2. Select a button with class `submit-btn`.
3. Select the 2nd `<li>` inside a `<ul>` with class `nav`.
4. Select any `<input>` whose `placeholder` contains the word "email".
5. Select a link (`<a>`) inside a `<footer>` tag.